

CLINICAL TRIALS AND OBSERVATIONS

Venetoclax plus pediatric regimen in adolescents and adults with Ph-negative acute lymphoblastic leukemia

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KEY POINTS

- Venetoclax plus pediatric-inspired chemotherapy significantly improved MRD negativity and survival in ND Ph⁻ ALL.
- The combination maintained a manageable toxicity profile.

The B-cell lymphoma 2 inhibitor venetoclax has shown promise in acute lymphoblastic leukemia (ALL), but its role in first-line therapy for newly diagnosed (ND) Philadelphia chromosome-negative (Ph⁻) ALL is undefined. In this prospective phase 2 study, 167 adolescents and adults (aged 14-60 years) with ND Ph⁻ ALL received venetoclax combined with pediatric-inspired chemotherapy. The primary end point was the rate of measurable residual disease (MRD) negativity by multiparameter flow cytometry (MFC) after induction. The complete remission rate was 91.0%, and 73.0% of responders achieved MFC-MRD negativity, meeting the primary end point. After a median follow-up of 19.3 months, median overall and disease-free survival were not reached; estimated 2-year survival rates were 78.5% and 76.7%, respectively. Propensity score-matched analysis confirmed superior survival compared with historical chemotherapy-only controls. Grade ≥ 3 adverse events were primarily hematologic toxicities and infections, with

an incidence comparable with that of the historical cohort. These results demonstrate that adding venetoclax to pediatric-inspired chemotherapy significantly improves MRD response and survival outcomes in newly diagnosed (ND) Ph⁻ ALL, with a manageable safety profile. This trial was registered at www.clinicaltrials.gov as #NCT05660473.

Introduction

Philadelphia chromosome-negative acute lymphoblastic leukemia (Ph⁻ ALL) represents a highly aggressive hematologic malignancy, characterized by the clonal expansion of immature lymphoid cells within the bone marrow (BM), peripheral blood, and extramedullary sites. Despite significant advancements in risk-adapted therapeutic strategies that have markedly enhanced survival rates in pediatric populations, outcomes for adolescent and adult patients with Ph⁻ ALL remain suboptimal. These groups experience higher relapse rates and reduced long-term survival compared with pediatric patients.^{1,2} Over the last 2 decades, the adoption of pediatric-inspired chemotherapy regimens has become increasingly prevalent as the standard induction treatment for fit adolescents and adults with Ph⁻ ALL. This shift has led to improved survival outcomes compared with traditional adult chemotherapy protocols.³⁻⁹

The persistence or recurrence of measurable residual disease (MRD) in ALL is well established as the most significant risk

factor for hematologic relapse and indicates resistance to standard chemotherapy.^{10,11} However, a consensus regarding the optimal MRD time point for predicting survival outcomes in adult ALL has not yet been reached. Several studies, along with our own research, support the independent prognostic value of postinduction MRD for survival outcomes.^{5,8,12} In adult patient cohorts, the rate of MRD negativity, as determined by multiparameter flow cytometry (MFC) after standard induction with pediatric-inspired chemotherapy, is ~50%.^{5,8,9,13} This statistic underscores a significant unmet need to enhance the depth of remission. Consequently, the optimization of induction regimens, with the objective of increasing both remission rates and MRD negativity beyond the capabilities of current pediatric-inspired protocols, has become a priority in contemporary research on Ph⁻ ALL.

Venetoclax, a potent and selective inhibitor of the anti-apoptotic protein B-cell lymphoma 2, has exhibited promising antileukemic activity across various lymphoid malignancies, including precursor B- and T-cell ALL, in preclinical studies.¹⁴⁻¹⁷

The combination of venetoclax with conventional chemotherapy regimens has demonstrated encouraging efficacy in patients with relapsed or refractory ALL.^{18,19} These findings provide a strong rationale for exploring venetoclax-based combinations in the frontline setting for Ph⁻ ALL.²⁰⁻²² Based on preclinical and early clinical observations, we conducted a prospective study to evaluate the efficacy and safety of venetoclax in combination with a pediatric-inspired chemotherapy regimen in adolescents and adults with newly diagnosed (ND) Ph⁻ ALL.

Methods

Study design and participants

This investigator-initiated, single-center, single-arm phase 2 trial (ClinicalTrials.gov identifier: NCT05660473) enrolled patients aged 14 to 60 years with ND Ph⁻ ALL after obtaining written informed consent. The study protocol was reviewed and approved by the Blood Diseases Hospital Ethics Committee, ensuring compliance with institutional ethical standards and the principles outlined in the Declaration of Helsinki. The primary end point of this study was the negativity rate of MRD assessed by MFC after induction chemotherapy, whereas the secondary end points included overall survival (OS), disease-free survival (DFS), cumulative incidence of relapse (CIR), and safety.

Treatment protocol

All enrolled patients received treatment according to the Institute of Hematology and Blood Diseases Hospital (IH)-2022 protocol, with detailed regimen specifications provided in supplemental Table 1, available on the *Blood* website. The IH-2022 protocol was developed by integrating venetoclax into the framework of our center's established pediatric-inspired protocol (IH-2014),⁸ which was itself a single-center, single-arm, prospective cohort study, with concomitant dose adjustments to the conventional chemotherapeutic agents. The induction therapy regimen includes vincristine at 1.4 mg/m² (capped at a maximum dose of 2 mg) administered on days 1, 8, 15, and 22; daunorubicin at 30 mg/m² on days 1 to 3; cyclophosphamide at 1200 mg/m² on days 1 and 15; pegylated asparaginase at 2500 IU/m² (capped at a maximum dose of 3750 IU) on day 5; prednisone at 1 mg/kg from days 1 to 14, and 0.5 mg/kg from days 15 to 28; and venetoclax at 100 mg on day 6, 200 mg on day 7, and 400 mg from days 8 to 14. On day 14 of the induction phase, all patients underwent BM aspiration. Patients exhibiting ≥10% residual leukemic blasts in the BM on day 14 of induction therapy were administered an additional 7-day course of venetoclax at a daily dose of 400 mg. After the observation of delayed hematopoietic recovery in 2 cases with prolonged venetoclax exposure, the protocol was revised to incorporate a fixed-duration administration of venetoclax (400 mg/day for 7 days during induction therapy). The revised protocol was implemented starting from the 35th enrolled patient, thereby eliminating the response-adaptive treatment extension based on the day-14 BM assessment. The induction treatment scheme is detailed in supplemental Figure 1. The comparison of induction-phase medications between the IH-2014 and IH-2022 regimens is provided in supplemental Table 2.

After induction, patients achieving complete remission (CR) entered consolidation therapy, whereas those without CR received the same regimen as salvage therapy. In both scenarios, all consolidation/salvage cycles included a 7-day augmentation with venetoclax (400 mg/day). Patients who failed to achieve CR after 2 treatment courses were withdrawn from the protocol. During the maintenance therapy phase, venetoclax is administered at 400 mg daily for a duration of 1 week, every 3 months. For patients diagnosed with BCR::ABL1-like ALL exhibiting genetic aberrations involving ABL-class kinases (eg, ABL1, ABL2, PDGFRB, CSF1R fusions), the treatment protocol incorporates the concurrent administration of a tyrosine kinase inhibitor (eg, dasatinib) alongside standard intensive chemotherapy. During the consolidation phase, patients with B-cell ALL (B-ALL) who have sufficient financial resources may receive 1 to 4 cycles of blinatumomab in addition to standard chemotherapy. However, this is not a standard treatment option, because only 20.6% of patients with B-ALL ultimately included blinatumomab in their treatment regimen. Prophylaxis for central nervous system (CNS) leukemia was conducted in accordance with the IH-2014 protocol.⁸

Response evaluation and MRD analysis

The criteria for treatment response and the methodology for MRD detection are detailed in the supplemental Definitions and MRD Assessment. MRD levels were monitored longitudinally throughout treatment using 2 complementary methodologies: MFC, with a sensitivity of 10⁻⁴; and next-generation sequencing (NGS)-based immunoglobulin/T-cell receptor sequencing, with a sensitivity of 10⁻⁶. MRD levels determined by MFC were classified as negative if <0.01%, and positive if ≥0.01%. In contrast, MRD status via NGS was categorized as undetectable (no sequencing reads identified), negative but detectable (<1 × 10⁻⁶ and above the limit of detection), and positive (MRD level of ≥1 × 10⁻⁶).

Risk stratification and allogeneic transplantation indications

According to international guideline²³ and our previous research,⁸ patients were designated as being at high risk (HR) if they met any of the following criteria: (1) age of >40 years; (2) all cases of T-cell ALL (T-ALL); (3) failure to achieve CR or persistence of MFC-detectable MRD in patients with CR after induction therapy; or (4) B-ALL with adverse genetic features, including hypodiploidy (<44 chromosomes), low hypodiploidy (30-39 chromosomes), near triploidy (60-78 chromosomes), TP53 mutation, KMT2A rearranged, ZNF384 rearranged, MEF2D rearranged, BCR::ABL1-like ALL, and complex karyotype (≥5 chromosomal abnormalities). Patients without these HR features were categorized as having standard risk.

For patients at HR, allogeneic hematopoietic stem cell transplantation (allo-HSCT) is recommended as a consolidative treatment after CR. Patients at HR who are either ineligible for or opt out of allo-HSCT, along with patients with standard risk, receive consolidation chemotherapy and maintenance therapy.

Sample size calculation and statistical analysis

The sample size calculation was based on demonstrating a 15% improvement in the MRD negativity rate, as assessed by

MFC at the end of induction (EOI), compared with our historical control rate of 50%.⁸ Considering an anticipated 20% dropout rate, a minimum of 102 patients was necessary to achieve 80% statistical power to detect a >20% difference from the historical control, using a 1-sided significance level of $\alpha = .05$. To ensure adequate power for subgroup analyses (particularly in T-ALL), enrollment continued beyond this minimum, resulting in a final cohort of 167 patients.

Statistical analyses were conducted using GraphPad Prism 9.0 (GraphPad Software, San Diego, CA) and R 4.0 (R Foundation for Statistical Computing, Vienna, Austria). Nonnormally distributed continuous variables, expressed as median (range), were analyzed using the Mann-Whitney *U* test. Categorical variables were compared by the χ^2 test or the Fisher exact test, as appropriate. Kaplan-Meier estimates were used for survival analysis, and differences between groups were evaluated using the log-rank test. The CIR was assessed using competing risks regression, with significance determined by the Gray method. The data cutoff for survival analysis was 31 March 2025. To ensure comparability of baseline characteristics between the IH-2022 and IH-2014 cohorts, propensity score matching (PSM) was used. Propensity scores were calculated using a logistic regression model, and a 1:1 nearest-neighbor matching algorithm with a caliper of 0.02 was applied for the comparative outcome analysis.

The study was approved by the ethics committee of the Institute of Hematology and Blood Diseases Hospital. Informed consents were obtained for this study and the publication of the manuscript.

Results

Patient characteristics and disposition

A cohort of 167 patients with Ph⁻ ALL was prospectively enrolled from June 2022 to May 2024. The demographic and baseline clinical characteristics of the study population are detailed in Table 1. The median age of the participants was 31 years (range, 14-60), with T-ALL comprising 24.6% of the cases. In accordance with the International Consensus Classification of B-ALL,²⁴ the genetic and molecular subtypes of the 126 patients with B-ALL are comprehensively depicted in supplemental Figure 2. It is noteworthy that the *ZNF384* rearrangement emerged as the most prevalent molecular aberration. Within the T-ALL subtypes, early T-cell precursor ALL (ETP-ALL) constituted 36.6% of the cases. The disposition of patients is illustrated in Figure 1.

Remission induction

All participants in the study received induction therapy. During this phase, 7 patients succumbed, resulting in an induction-related mortality rate of 4.19%. Of 160 patients assessed for efficacy, 152 (152/167 [91.0%]) achieved CR after induction chemotherapy. Among 8 patients who did not respond to induction therapy, only 1 (1/8 [12.5%]) attained CR after the first salvage chemotherapy regimen. The comprehensive clinical characteristics of patients who did not achieve CR after induction chemotherapy are detailed in supplemental Table 3. After 2 courses of chemotherapy, the overall CR rate was 91.6% (153/167).

Table 1. Patient baseline characteristics

Characteristics	(N = 167)
Sex, n (%)	
Male	97 (58.8)
Female	70 (41.2)
Age at diagnosis, y	
Median	31
Range	14-60
<40 y, n (%)	115 (68.9)
≥40 y, n (%)	52 (31.1)
WBC, ×10⁹/L	
Median	10.64
Range	1.06-410.00
ALL subset, n (%)	
B-lineage ALL	N = 126
<i>ZNF384</i> rearrangement	22 (17.5)
<i>BCR::ABL1</i> -like	13 (10.3)
t(v;11q23.3)/ <i>KMT2A</i> rearrangement	11 (8.7)
t(1;19)(q23.3;p13.3)/ <i>TCF3::PBX1</i>	11 (8.7)
Hyperdiploid	8 (6.3)
Low hypodiploid	6 (4.8)
<i>MEF2D</i> rearrangement	3 (2.4)
<i>DUX4</i> rearrangement	3 (2.4)
<i>PAX5</i> P80R	3 (2.4)
t(12;21)(p13.2;q22.1)/ <i>ETV6::RUNX1</i>	2 (1.6)
<i>IKZF1</i> N159Y	1 (0.8)
<i>PAX5</i> alteration	1 (0.8)
B-ALL, NOS	42 (33.3)
T-lineage ALL	N = 41
ETP-ALL	15 (36.6)
Non-ETP ALL	26 (63.4)

NOS, not otherwise specified; WBC, white blood cell.

OS, DFS, and CIR

After a median follow-up period of 19.3 months, the median OS for the entire cohort and the median DFS for patients who achieved CR within 2 treatment cycles had not yet been reached. The estimated 2-year OS and DFS rates were 78.5% (95% confidence interval [CI], 71.8-85.2) and 76.7% (95% CI, 69.6-83.8), respectively (Figure 2A). With patients censored at the time of allo-HSCT, the estimated 2-year OS and DFS rates were 79.3% (95% CI, 70.3-88.3) and 79.7% (95% CI, 70.9-79.8), respectively (Figure 2B). Table 2 shows the estimated 2-year OS and DFS rates of the overall cohort and subgroups. Comparative survival analyses across subgroups are illustrated in supplemental Figure 3. Among patients who achieved CR, 21 experienced relapses by the end of follow-up. The 2-year CIR was 15.0% (95% CI, 9.0-21.1). Of these, 10 relapses occurred during the initial chemotherapy phase, whereas 11 occurred after undergoing allo-HSCT in their first CR. The sites of first relapse were as follows: isolated BM in 16 patients, isolated CNS in 2, and combined BM and CNS in 3.

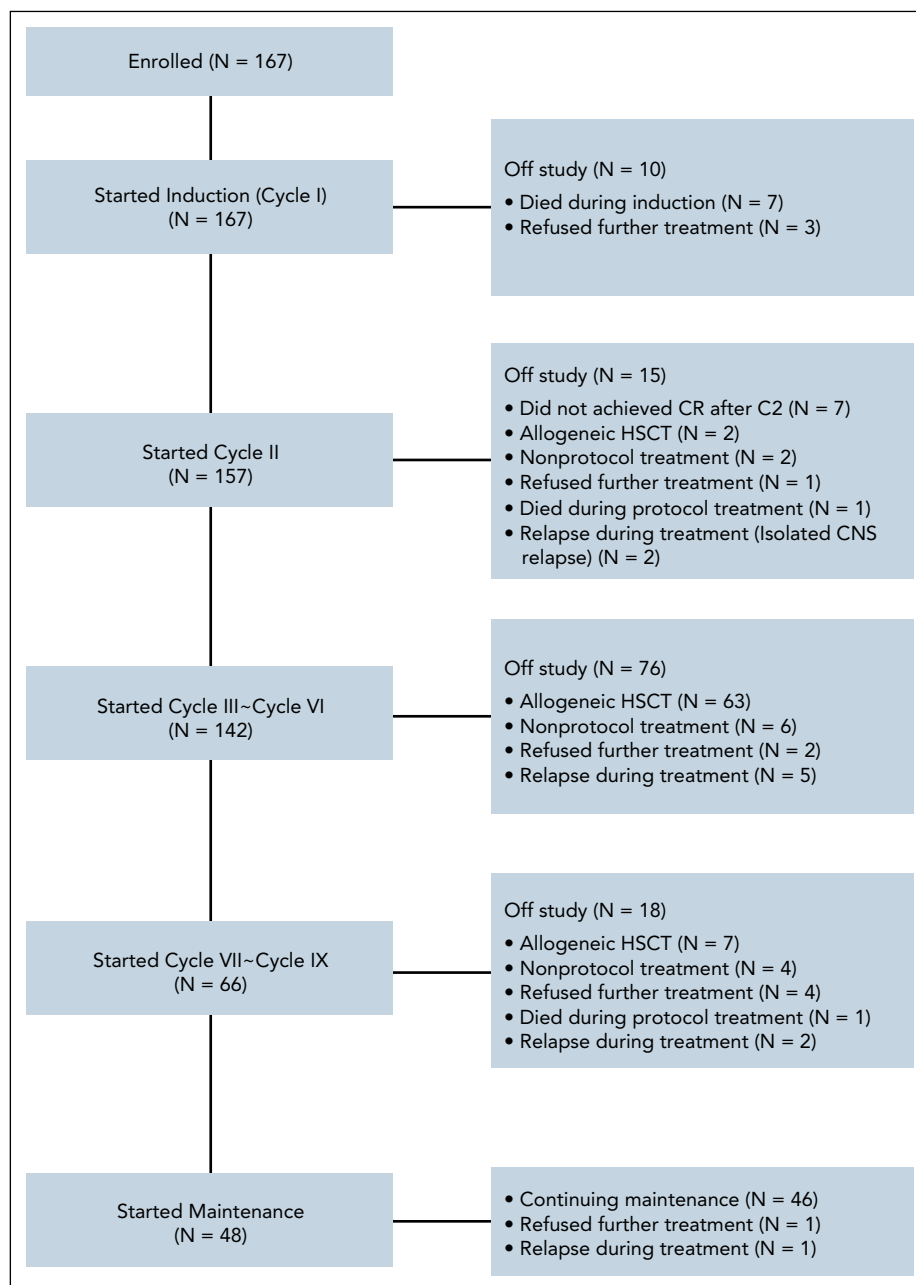


Figure 1. Flowchart of patient disposition.

MRD results

A total of 157, 152, and 135 BM samples were available for MRD assessment by MFC on day 14 of induction, at EOI, and after the first consolidation cycle (designated post-C1 and corresponding to cycle 2 in supplemental Table 1), respectively. In addition, NGS-MRD data were available for 80 samples at both the EOI and post-C1 time points. The analysis at the EOI and post-C1 time points was restricted to patients who had achieved CR. Figure 3A-B illustrates the longitudinal dynamics of MRD status as assessed by MFC and NGS, respectively. Figure 3C-D further details the composition of MRD levels at the EOI, stratified by disease subtype.

The MFC-MRD negativity rate after induction therapy reached 73.0% (111/152; 95% CI, 65.5-79.6) among patients who achieved CR (Figure 3A). Because the lower bound of the 95% CI (65.5%) exceeded the prespecified target threshold of 15%, the primary end point of the trial was met. The MFC-MRD negativity rate was consistent at 73.0% for both B-ALL and T-ALL, with no statistically significant difference between these subtypes. In ETP-ALL, the MFC-MRD negativity rate was 71.4%, which was comparable with that in non-ETP ALL (73.9%). Notably, within the B-ALL subtype, the genetic subgroups *BCR::ABL1*-like ALL and *ZNF384* rearrangement exhibited suboptimal MRD clearance as assessed by MFC. In

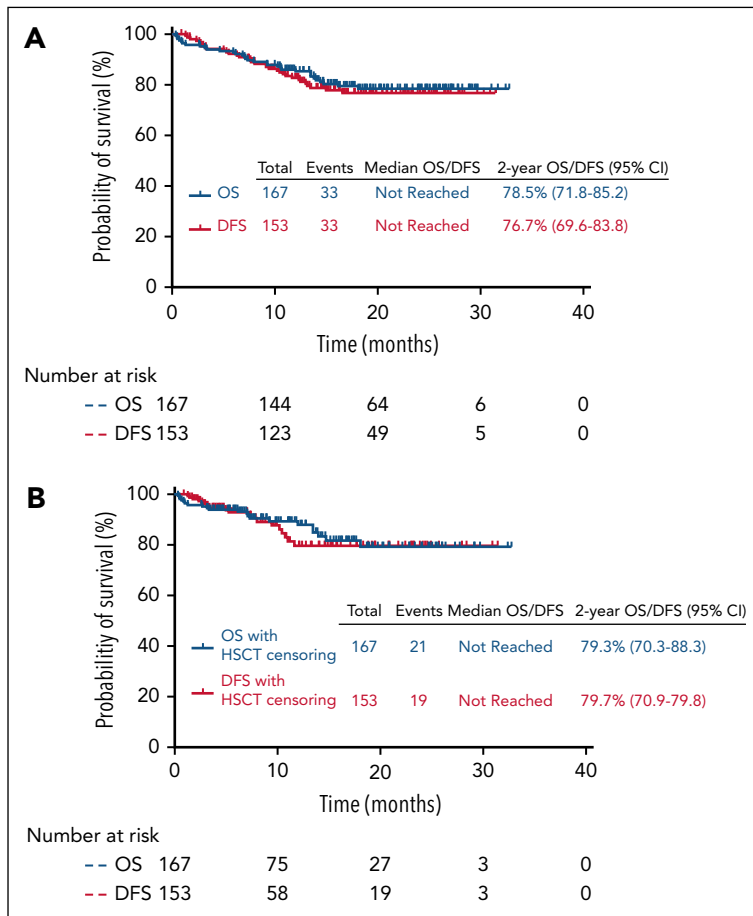


Figure 2. Survival outcomes of the study cohort. (A) OS and DFS for the entire cohort. (B) OS and DFS with patients censored at the time of allo-HSCT.

contrast, the MFC-MRD negativity rate approached 100% in other B-ALL subtypes (Figure 3C). The disposition for patients with CR who were MFC-MRD-positive after induction is presented in supplemental Figure 4.

The rates of undetectable MRD as determined by NGS were 26.3% after induction therapy and 40.0% post-C1, respectively (Figure 3B). Using a threshold of $<10^{-6}$ to define NGS-MRD negativity, 36.3% of patients achieved NGS-MRD negativity after induction therapy, which increased to 62.5% after the post-C1 assessment. Matched-pair analyses for MFC-MRD and NGS-MRD were conducted on 80 patients who achieved CR after induction. The distribution of MRD levels, as measured by each method across various disease subtypes, is depicted in supplemental Figure 5.

Impact of MRD status on survival

The prognostic significance of MRD varied based on the timing of assessment. At EOI, MRD status, whether determined by MFC or NGS, did not exhibit a statistically significant correlation with 2-year outcomes. For patients stratified by MFC-MRD at EOI, the estimated 2-year OS rates for patients with negative MRD vs positive MRD was 84.0% vs 79.7% ($P = .670$), DFS was 78.1% vs 73.1% ($P = .561$), and CIR was 12.1% vs 21.9% ($P = .175$; Figure 4A-C). Similarly, outcomes were comparable between patients with negative NGS-MRD and positive NGS-MRD at this time point, with no significant differences

observed in 2-year OS (93.1% vs 84.1%; $P = .390$), DFS (81.1% vs 72.9%; $P = .639$), or CIR (12.0% vs 23.1%; $P = .367$; Figure 4D-F).

In contrast, the post-C1 assessment of MRD demonstrated significant prognostic value for relapse risk but not for OS. Among patients stratified by post-C1 multiparameter MFC-MRD status, the estimated 2-year OS rates for patients with negative MRD and positive MRD were 85.3% and 75.2% ($P = .230$), respectively. However, DFS was significantly reduced in patients with positive MRD (80.8% vs 59.8%; $P = .029$), and the CIR was significantly elevated (11.8% vs 34.6%; $P = .007$) in patients with positive MRD (Figure 5A-C). A similar trend was observed with NGS-MRD, with which OS did not differ significantly between patients with negative MRD and positive MRD (92.0% vs 81.8%; $P = .310$), yet patients with positive MRD exhibited inferior DFS (89.1% vs 61.8%; $P = .015$) and a higher CIR (4.9% vs 34.7%; $P = .002$; Figure 5D-F).

Univariate and multivariate analyses of DFS and OS

Supplemental Table 4 shows the results of the univariate analysis of factors associated with DFS and OS. Given the low number of events ($n = 19$ for DFS; $n = 33$ for OS), which is insufficient for a reliable multivariable Cox regression model according to standard statistical power considerations, we refrained from performing such an analysis to prevent

Table 2. Estimated 2-year OS and DFS rates of the overall cohort and subgroups

Types of patients	2-year OS			2-year DFS		
	n	%	95% CI	n	%	95% CI
Overall	167	78.5	71.8-85.2	153	76.7	69.6-83.8
B-ALL	126	80.3	73.0-87.6	115	75.6	67.0-84.2
<i>ZNF384</i> rearrangement	22	80.6	63.4-97.8	21	75.2	56.2-94.2
<i>BCR::ABL1</i> -like	13	78.6	51.4-100.0	13	63.5	34.5-92.5
<i>t</i> (v;11q23.3)/ <i>KMT2A</i> rearrangement	11	75.0	45.0-100.0	11	74.1	42.5-100.0
<i>t</i> (1;19)(q23.3;p13.3)/ <i>TCF3::PBX1</i>	11	80.0	55.3-100.0	11	70.7	42.7-98.7
Hyperdiploid	8	87.5	64.6-100.0	7	66.7	13.4-100.0
Low hypodiploid	6	33.3	0-70.9	3	66.7	13.4-100.0
B-ALL, NOS	42	80.7	68.5-92.9	39	77.4	63.5-91.3
B-ALL with blinatumomab incorporation	26	82.7	67.2-98.4	26	75.0	57.4-92.6
B-ALL without blinatumomab incorporation	100	79.7	71.5-87.9	92	75.6	65.8-85.4
MFC-MRD–positive B-ALL with blinatumomab incorporation	6	66.7	29.1-100.0	6	50.0	10.0-90.0
MFC-MRD–negative B-ALL with blinatumomab incorporation	20	89.1	74.8-100.0	20	84.4	68.1-100.0
T-ALL	41	73.6	59.3-87.9	38	79.2	66.3-92.1
ETP-ALL	15	78.8	57.2-100.0	15	80.0	59.8-100.0
Non-ETP T-ALL	26	70.5	51.7-89.3	23	78.7	62.0-95.4
Age <40 years	115	86.2	79.3-93.1	110	80.8	73.0-88.6
Age ≥40 years	52	61.7	48.0-75.4	43	67.1	52.8-81.4

NOS, not otherwise specified.

overfitting. Therefore, in this study, univariate survival analyses were conducted using the Kaplan-Meier method, with log-rank tests as the primary strategy to evaluate the prognostic significance of each variable.

Propensity score analysis

To ensure the comparability of baseline characteristics between patients from the IH-2022 and IH-2014 cohorts, the propensity score model incorporated clinically relevant variables, including sex, age, white blood cell count at diagnosis, serum lactate dehydrogenase level at diagnosis, and immunophenotype (classified as T-ALL or B-ALL). The baseline characteristics of the 2 balanced cohorts after matching are summarized in Table 3. The 2-year OS and DFS rates were 78.2% vs 58.1% ($P = .001$) and 76.4% vs 58.2% ($P = .002$), respectively, with both comparisons demonstrating statistically significant differences (Figure 6A-B). Subgroup analyses further revealed statistically significant improvements in both OS and DFS for the IH-2022 cohort compared with the IH-2014 cohort, which were consistently observed in both the B-ALL (2-year OS rates: 80.3% vs 62.1%, $P = .007$; 2-year DFS rates: 75.5% vs 61.0%, $P = .017$) and T-ALL (2-year OS rates: 72.6% vs 44.4%, $P = .035$; 2-year DFS rates: 78.1% vs 46.6%, $P = .021$) immunophenotype subgroups (Figure 6C-F). To mitigate potential

confounding from blinatumomab use ($n = 26$), a post hoc sensitivity analysis excluding these patients was conducted. The results remained consistent with the primary analysis, with the venetoclax-containing regimen maintaining a significant survival advantage (2-year OS rates: 79.7% vs 62.1%, $P = .023$; 2-year DFS rates: 75.5% vs 61.0%, $P = .022$). It should be emphasized that these results are exploratory, post hoc subgroup analyses. The PSM was performed for the overall cohort and was not specifically calibrated to achieve balance within the B-ALL and T-ALL subgroups.

Treatment-related toxicities

During the induction period, 7 early deaths occurred in the IH-2022 cohort, resulting in an early mortality rate of 4.19% (7/167). The median age of these deceased patients was 46 years, which was significantly higher than the cohort-wide median age of 31 years. The primary cause of death was pulmonary infection, accounting for 3 cases. Additional fatal events included bacterial sepsis ($n = 1$), candidemia ($n = 1$), and cerebrovascular accidents ($n = 2$). After induction, 2 further deaths due to treatment-related infections occurred during the consolidation phase. When compared with the IH-2014 protocol, which had an early mortality rate of 2.89% (12/415), the

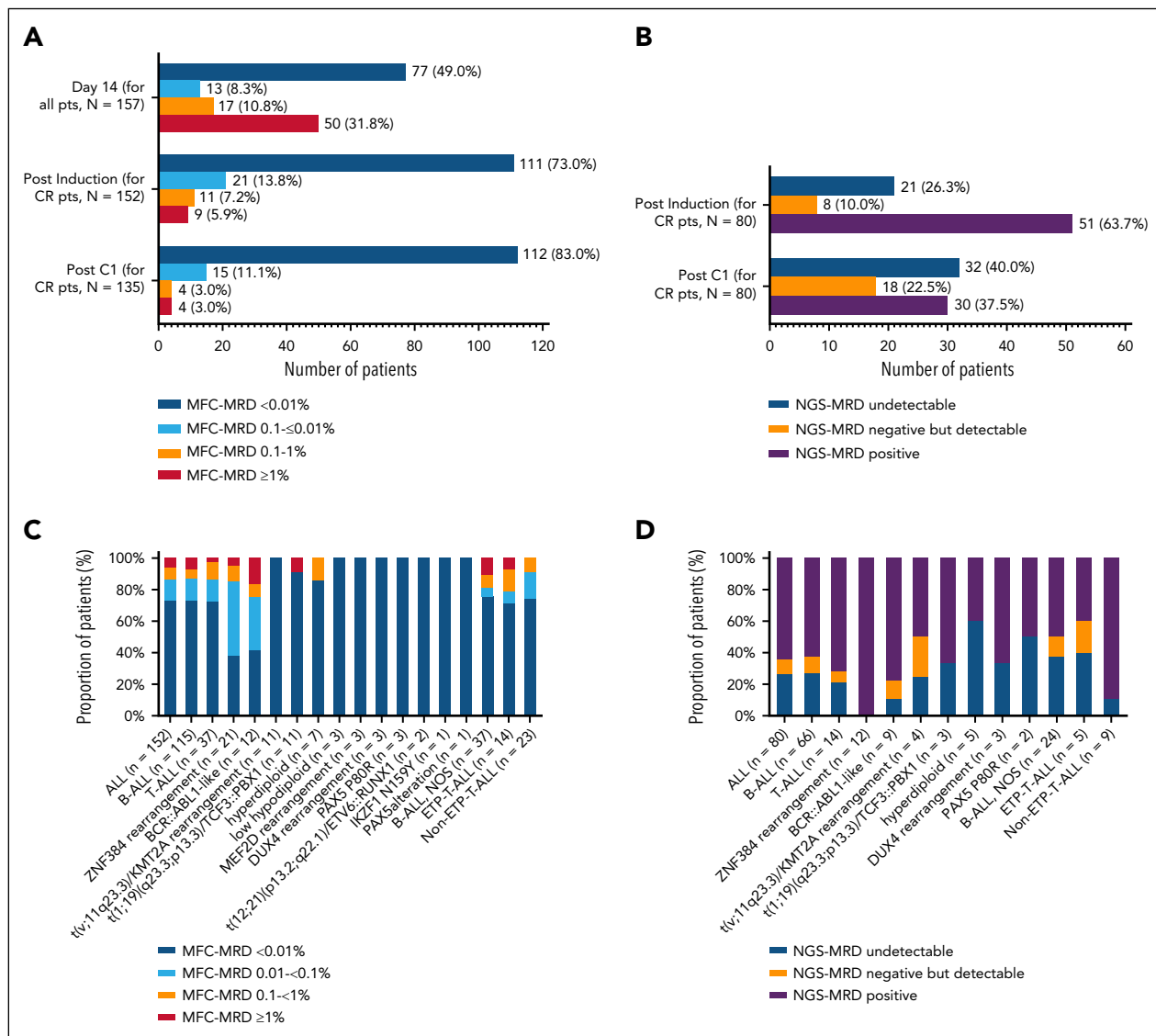


Figure 3. Dynamics and composition MRD. (A-B) Longitudinal monitoring of MRD levels assessed MFC at 3 time points (day 14 of induction, after induction, and after the first consolidation post-C1) (A), and by NGS at 2 time points (after induction and post-C1) (B). (C-D) Distribution of MRD levels after induction, stratified by disease subtype, as determined by MFC (C) and NGS (D). NOS, not otherwise specified; pts, patients.

difference between the 2 cohorts was not statistically significant ($P = .430$).

In this cohort, following the IH-2022 protocol, the median duration of grade 4 neutropenia (absolute neutrophil count of $0.5 \times 10^9/L$) during induction therapy was 18 days (range, 4-37), and the median duration of grade ≥ 3 thrombocytopenia (platelet count of $<50 \times 10^9/L$) was 14 days (range, 0-43). These durations were comparable with those observed in the historical control cohort following the IH-2014 protocol, which were 19 days (range, 0-43) for grade 4 neutropenia and 15 days (range, 0-45) for grade ≥ 3 thrombocytopenia, with no statistically significant differences between the 2 cohorts (supplemental Figure 6A-B). In patients who achieved CR and advanced to the first cycle of consolidation chemotherapy, no statistically significant differences were observed between the IH-2022 and IH-2014 protocols in terms of the median duration of grade 4 neutropenia (9 days [range, 0-39] vs 8 days [range,

0-17], respectively) or grade ≥ 3 thrombocytopenia (8 days [range, 0-45] vs 9 days [range, 0-21], respectively; supplemental Figure 6C-D). Although the median durations did not differ statistically, it is noteworthy that the maximum durations of both grade 4 neutropenia and grade ≥ 3 thrombocytopenia were longer in the IH-2022 cohort than in the IH-2014 cohort during consolidation therapy, indicating that a subset of patients experienced exceptionally prolonged myelosuppression under the newer protocol.

Concerning serious infectious complications during the induction phase, the incidence of grade ≥ 3 events, such as bloodstream infections (15.0% vs 19.4%; $\chi^2 = 1.076$; $P = .300$) and pneumonia (34.4% vs 26.3%; $\chi^2 = 2.500$; $P = .114$), was comparable between the IH-2022 and IH-2014 protocols, with no statistically significant differences detected. Notably, however, a distinct and significant disparity emerged in the rate of noninfectious severe toxicity. The incidence of grade ≥ 3

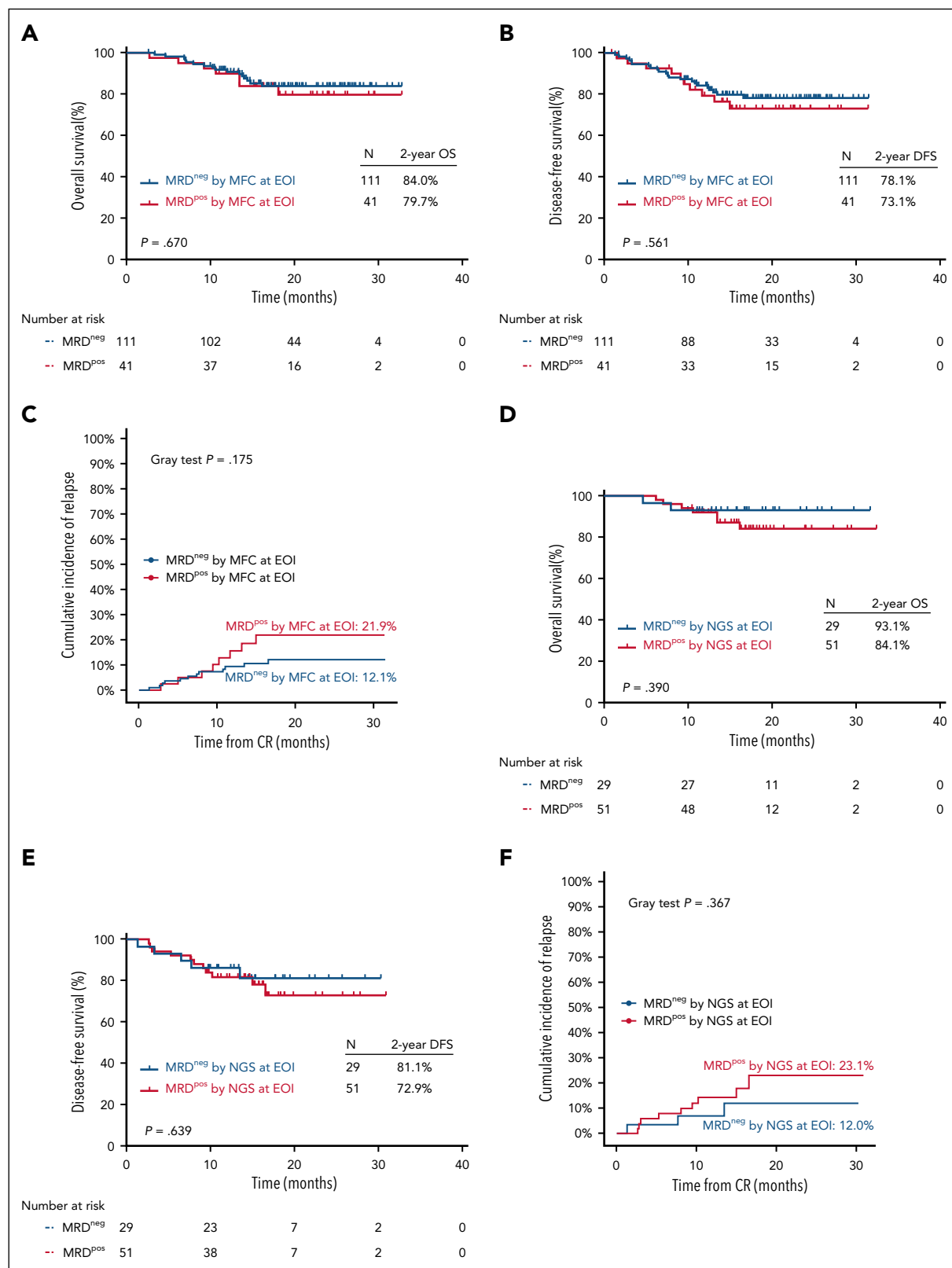


Figure 4. Impact of MRD status on survival outcomes after induction therapy. (A-C) Curves showing OS (A), DFS (B), and CIR (C), stratified by MRD status as determined by MFC at the EOI. (D-F) Corresponding analyses for MRD status assessed by NGS at EOI: OS (D), DFS (E), and CIR (F).

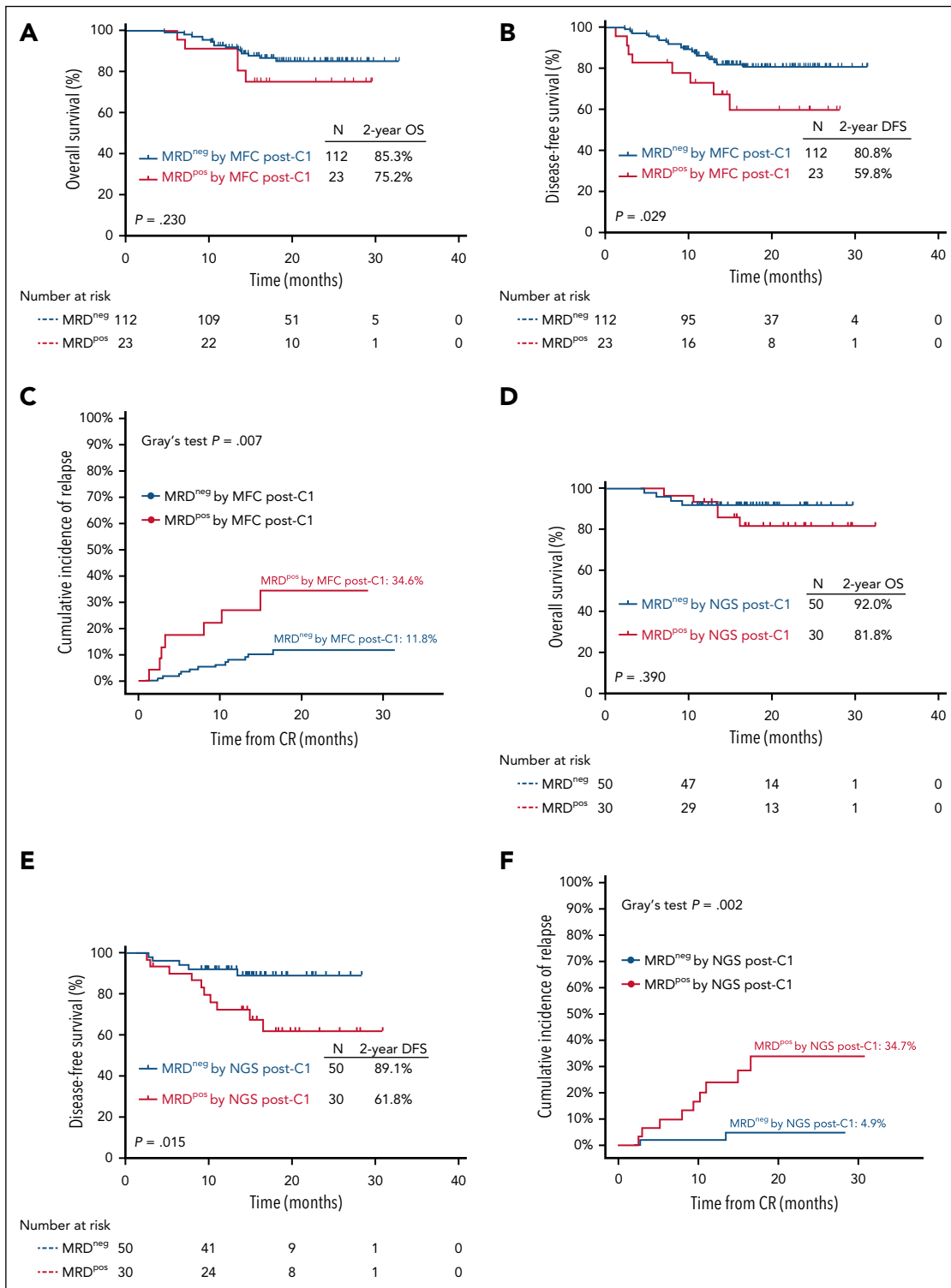


Figure 5. Impact of MRD status on survival outcomes after the first consolidation. (A-C) Curves showing OS (A), DFS (B), and CIR (C), stratified by MRD status assessed by MFC outcomes after the first consolidation cycle. (D-F) Corresponding analyses for MRD status assessed by NGS after the first consolidation cycle: OS (D), DFS (E), and CIR (F).

intestinal obstruction was markedly higher in the IH-2022 protocol than in the IH-2014 protocol (18.1% vs 5.6%; $\chi^2 = 11.945$; $P < .001$). Supplemental Table 5 provides a comparison of the frequency of common grade ≥ 3 adverse events between the 2 cohorts during the induction phase.

Discussion

This study demonstrates that the incorporation of venetoclax into a pediatric-inspired chemotherapy regimen significantly enhances outcomes in adolescents and adults with ND Ph⁺ ALL. The primary

Table 3. Baseline characteristics of the propensity score-matched cohorts

Characteristics	IH-2022 (N = 160)	IH-2014 (N = 160)	P value
Male sex, n (%)	95 (59.4)	95 (59.4)	1.000
Age, median (range), y	31 (14-60)	29 (14-60)	.219
WBC count, median (range), $\times 10^9/L$	10.14 (1.06-410.00)	11.93 (0.73-504.00)	.576
LDH, median (range), IU/L	444 (24-9774)	515 (75-7073)	.219
Immunophenotype, B-ALL, n (%)	121 (75.6)	124 (77.5)	.692
Allo-HSCT in CR1, n (%)	75 (46.9)	65 (40.6)	.260

CR1, first CR; LDH, lactate dehydrogenase; WBC, white blood cell.

end point was met, with the rate of MFC-MRD negativity after induction therapy increasing markedly from 50% in the historical cohort to 73% in this study. Furthermore, in comparison with a historical cohort treated with chemotherapy alone (IH-2014), the venetoclax combination regimen was associated with superior DFS and OS in both B-ALL and T-ALL, highlighting its broad efficacy across different immunophenotypes.

Patients diagnosed with ETP-ALL, a subtype traditionally linked with poor responsiveness to standard chemotherapy, demonstrated notably significant benefits. The treatment regimen resulted in substantially elevated rates of CR and MRD eradication within this HR cohort, with postinduction MFC-MRD negativity increasing from 19.4% to 71.4%.⁸ These findings indicate that venetoclax may effectively counteract the inherent chemoresistance associated with ETP-ALL. This discovery holds considerable clinical significance because it presents a promising therapeutic avenue for a patient group with previously limited effective treatment options. Nevertheless, the responses were not consistent across all genetic subtypes. In B-ALL, patients with *BCR::ABL1*-like ALL or those harboring *ZNF384* rearrangements derived less benefit in terms of MRD clearance. This variance highlights the molecular heterogeneity of B-ALL and suggests that intrinsic or acquired resistance mechanisms to venetoclax may be operative in these subsets. Our clinical data revealed a paradoxical discrepancy between in vitro biomarker profiling and therapeutic outcomes in *BCR::ABL1*-like ALL. Despite preclinical BH3 profiling suggesting a higher degree of B-cell lymphoma 2 dependence in *BCR::ABL1*-like ALL blasts than non-*BCR::ABL1*-like counterparts,²⁰ our cohort demonstrated a suboptimal response to the venetoclax combination regimen within this genetic subtype. This finding underscores the necessity for further investigation into the molecular determinants of venetoclax sensitivity in these subtypes, which may inform the development of combination strategies with other targeted agents.

In this prospective cohort study, we observed a notably high rate of MRD negativity assessed by MFC after induction therapy (73.0%), which significantly exceeded the historical benchmark of 50% used for sample size calculation. Contrary to our initial hypothesis and previous evidence,^{8,11} however, this early MRD clearance did not correlate with improved survival outcomes. We hypothesize that this discrepancy may be attributed to a ceiling effect: when a substantial proportion of patients achieve a favorable response, the binary

MRD-negative status at this early time point loses its prognostic discriminatory power. Patients who remain MRD positive after the initial consolidation cycle (post-C1) likely constitute a subgroup with highly resistant disease, potentially serving as a more accurate predictor of relapse. Therefore, more sensitive biomarkers or later assessment time points are necessary to identify the residual HR population. Supporting this hypothesis, MRD status evaluated post-C1 has emerged as a robust predictor of relapse. However, this association does not correspond to a statistically significant difference in OS. This apparent disconnect between MRD status and OS may be attributed to 2 interrelated factors. Firstly, the relatively short follow-up period may be inadequate for survival differences to become fully apparent. Secondly, and more significantly, the high efficacy of contemporary salvage therapies, particularly immunotherapies and targeted agents such as blinatumomab, inotuzumab ozogamicin, and chimeric antigen receptor T-cell therapy, has the potential to successfully rescue a subset of patients who experience relapse, thereby reducing the influence of MRD status on long-term survival outcomes.²⁵ This observation highlights the evolving landscape of B-ALL treatment, in which advances in salvage strategies are progressively decoupling early relapse risk from overall mortality.

From a safety perspective, the incorporation of venetoclax into our pediatric-inspired chemotherapy backbone was generally well tolerated and did not lead to unexpected toxicities. Notably, in contrast to the 2 to 3 weeks of venetoclax administration associated with heightened myelotoxicity in similar protocols,²² our strategy of limiting treatment to 1 week per cycle achieved a superior therapeutic index; the myelosuppressive profile remained comparable with that of intensive chemotherapy alone, whereas clinical efficacy was significantly improved. The regimen, however, was associated with a significant incidence of severe gastrointestinal toxicity, specifically grade ≥ 3 intestinal obstruction, which exceeded initial expectations. This particular manifestation of gastrointestinal toxicity has not been prominently documented in previous studies involving venetoclax in combination with chemotherapy for ALL or acute myeloid leukemia.^{20,22,26,27} We hypothesize that this may be due to a synergistic effect on slowing intestinal motility between venetoclax and vincristine, both of which have been individually associated with gastrointestinal dysmotility. Further investigation is warranted to determine whether this is attributable to a synergistic interaction with our specific chemotherapy regimen, a population-specific susceptibility, or another

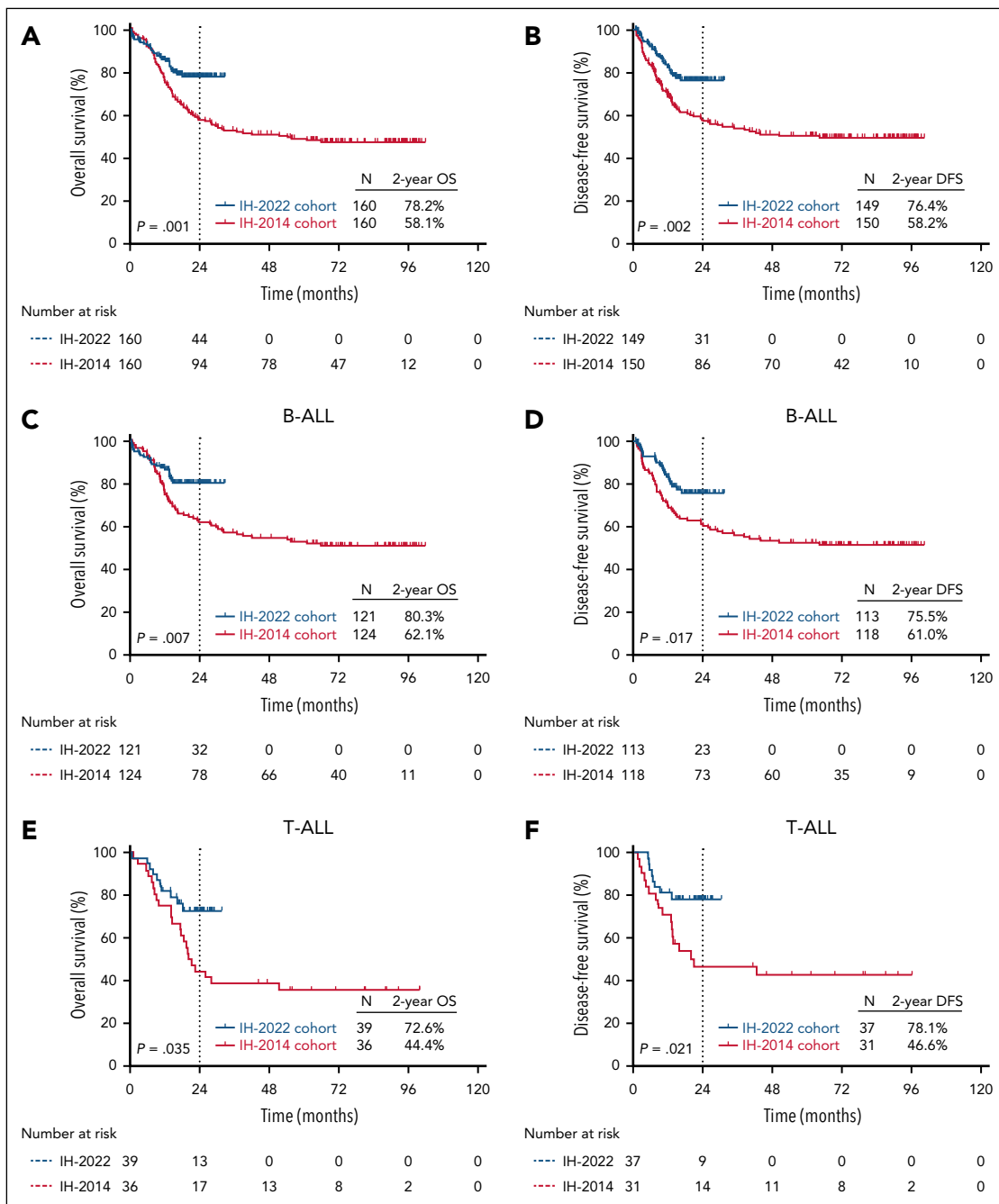


Figure 6. Comparison of survival outcomes between the IH-2022 and IH-2014 cohorts after PSM. (A-B) Kaplan-Meier curves for the entire matched cohorts: OS (A) and DFS (B). (C-D) Corresponding analyses for the B-ALL subgroup: OS (C) and DFS (D). (E-F) Corresponding analyses for the T-ALL subgroup: OS (E) and DFS (F).

underlying factor. As a result, and particularly during the induction phase when the risk may be highest, it is advisable to incorporate proactive preventive measures against gastrointestinal obstruction into the supportive care protocol. Such measures may include vigilant monitoring and the early administration of laxatives or prokinetics.

Our findings should be considered in the context of 2 primary limitations. First, the single-arm, nonrandomized design, despite using PSM against a historical control cohort, introduces potential biases. Importantly, the PSM comparison evaluates the bundled therapeutic advances of the

contemporary era (IH-2022) against a historical control (IH-2014). Therefore, the observed survival benefits, particularly within the B-ALL subgroup, should be interpreted as an outcome associated with the contemporary treatment package, which included venetoclax, refined transplantation strategies, and the selective use of blinatumomab, rather than being attributed exclusively to venetoclax. A randomized trial would be required to isolate the specific effect of venetoclax. Second, this analysis is limited by a short follow-up period and a small number of events, which may preclude the detection of statistically significant differences, especially for survival outcomes.

Conclusions

Our study illustrates that the incorporation of venetoclax into a pediatric-inspired chemotherapy framework significantly enhances the frontline treatment of Ph⁺ ALL, as evidenced by improved MRD negativity rates and survival outcomes. Nevertheless, this regimen remains an intensive chemotherapy strategy, characterized by notable treatment-related toxicity and the potential for fatal complications. This concern is amplified by the current paradigm shift in B-ALL therapy, for which novel immunotherapies and targeted antibodies are being successfully combined with reduced-intensity chemotherapy to improve efficacy while reducing toxicity.²⁵ In stark contrast, the frontline therapeutic landscape for T-ALL, particularly the ETP-ALL subtype, lacks effective novel agents. Consequently, we propose that the most promising and distinctive application for venetoclax may not be in further intensifying B-ALL regimens, but rather in addressing this critical gap in T-ALL/ETP-ALL. This hypothesis necessitates rigorous validation through dedicated randomized controlled trials to establish the role of venetoclax in this population with unmet needs. To this end, we are planning a multicenter randomized trial to prospectively compare the efficacy of our chemotherapy backbone with vs without venetoclax in this population.

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Authorship

Contribution: Y.M., H.W., and J.W. designed the clinical study protocol; X.G. analyzed data, interpreted results, and prepared the initial draft of the manuscript; and all authors executed the treatment protocol,

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Footnotes

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The online version of this article contains a data supplement.

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